District Police & Technical Cell Field Research Guide

Exhaustive Technical Questionnaire & Ground-Truth Assessment

Project: SHADOW-INTEL (AI Criminal Network Analysis)

Problem Statement ID: SIH26189

Sponsoring Ministry: Ministry of Home Affairs (MHA)

Supported Bodies: BPR&D / NCRB / I4C / NIA / State Police

> ***30-Second Icebreaker & Purpose:*** *"Sir/Ma'am, we are building an AI-powered Criminal Network Intelligence System for Smart India Hackathon (SIH26189 - Ministry of Home Affairs). We came directly to the SP Office and Technical Cell to understand your actual daily struggles with CDRs, Tower Dumps, Cyber Fraud, CCTNS, and Court Admissibility so our system solves genuine ground bottlenecks rather than theoretical problems."*

Section 1: Strategic & Organizational Perspective For SP / Addl. SP / DSP Crime

1. Syndicate Hierarchy & Insulation:

- What are the most prevalent organized crime categories in the district (e.g., Cyber fraud cartels, narcotics syndicates, inter-state burglary, extortion rings)?
- How many layers of operational insulation (mules, couriers, burner phone callers) typically separate the mastermind/kingpin from ground operatives?

2. Inter-District & Inter-State Data Sharing:

- When an organized gang operates across district or state borders, what is the exact mechanism for real-time intelligence exchange?
- How long does it currently take to discover that a suspect arrested locally is a wanted node in an inter-state syndicate?

3. Investigation Turnaround Times & Backlogs:

- What is the average turnaround time (days/weeks) to clean, process, and cross-correlate digital footprints (CDRs, bank accounts, forensic phone dumps) in a complex case?
- What percentage of an Investigating Officer's (IO) time is lost on manual spreadsheet correlation versus active field investigation?

4. Adoption & Trust in AI Decision Support:

- What is leadership's primary hesitation with AI tools (black-box decisions, false positives, legal non-admissibility, lack of technical training)?
- What specific mathematical evidence or plain-language reasoning must an AI system provide before leadership authorizes asset freezes or search warrants?

Section 2: Telecom, Location & Tower Dump Forensics For Technical Cell / Cyber Sub-Inspectors

5. Raw Telecom Formats (CDR / SDR / IPDR):

- What file formats do you receive from TSPs (Airtel, Jio, Vi, BSNL)? Are they Excel, CSV, text, or encrypted PDFs?
- Do different telcos have conflicting column headers, date-time formats, or time zones (IST vs. UTC)? How do you normalize them today?
- When verifying Subscriber Detail Records (SDR / CAF), how do you detect synthetic or forged Aadhaar/identity documents?

6. Burner Phones & IMEI-IMSI Hopping:

- How do syndicates rotate SIM cards and handsets (e.g., multiple SIMs in one phone, one SIM in multiple phones, cloned IMEIs like 00000000000000)?
- How do you currently track when a single suspect operates 5+ active numbers across different locations?

7. **Tower Dump Analytics at Crime Scenes:**

- When pulling a Tower Dump with 50,000 to 2,00,000 phone numbers active near a crime scene, what is your manual filtering workflow to isolate the top 2–3 suspect numbers?
- How do you filter out static resident numbers who permanently live under that cell tower's coverage?
- How do you cross-correlate multiple tower dumps from different crime scenes or highway toll escape routes?

8. **Common Link & Geo-Temporal Rendezvous Analysis:**

- When two suspects never call each other directly, how do you uncover their common middleman / broker node?
- How do you detect if two suspects met in person at a specific dhaba/hotel/checkpoint based purely on spatial-temporal cell-ID overlaps?

9. **VoIP, Virtual Numbers & OTT Bypass:**

- How frequently do criminals use WhatsApp/Signal calling, Telegram, or virtual numbers (+1, +44, +371) to bypass traditional telecom CDRs?
- When traditional CDR shows only IPDR (data sessions), how do you correlate that with suspect activity?

Section 3: Cyber Financial Fraud & Mule Account Ring Tracing For Cyber Crime IOs / 1930 Portal

10. **1930 / I4C / CFCFRMS Workflow:**

- How does a cyber fraud complaint reported on the National 1930 / I4C portal reach the district cyber cell?
- What is the typical time delay between fraud execution and the first bank account freeze order?

11. Mule Account Layering & Speed:

- In digital arrest, task fraud, and investment scams, how many layers (Layer-1 to Layer-5) does stolen money pass through before ATM cash withdrawal or crypto conversion?
- How do you map the flow of funds across disparate banking institutions (PSU banks, private banks, cooperative banks, payment wallets)?

12. Mule Syndicate Clustering:

- How do you identify whether 20 different bank accounts across different branches belong to the same local mule handler?
- What common attributes are most effective for linking (shared mobile number, same introducer, shared IP address during net-banking logins)?

13. Crypto & P2P Escrow Fraud:

- How often do fraud proceeds get converted into USDT or Bitcoin via P2P trading on crypto exchanges (Binance, Bybit)?
- What tools or methods do you currently possess to trace on-chain crypto wallet transactions? Where does the investigation hit a dead end?

Section 4: Multilingual Police Records & Modus Operandi NLP For Crime Branch & Investigating Officers

14. CCTNS / ICJS Ground Reality:

- How reliably can you search criminal records across other districts/states using CCTNS or ICJS?

- What percentage of FIR information is structured (fixed dropdowns) vs. unstructured (free-text narratives in regional language)?

15. **Multilingual Processing & Inter-State Language Barriers:**

- FIRs and case diaries are written in regional Indic languages (Hindi, Marathi, Tamil, Telugu, Bengali, Gujarati) or mixed Hinglish.
- If a criminal syndicate operates across state borders, how do you bridge the language barrier to match previous arrest records and aliases?

16. **Modus Operandi (MO) Semantic Search:**

- Can you currently search police databases by specific criminal methods (e.g., "theft of ATM using gas-cutters and white Bolero at night"), or only by exact suspect name?
- How valuable would a semantic AI search engine be that matches new unsolved FIRs against historical national crime patterns?

Section 5: Digital Forensics & Seized Device Intelligence For Forensic Lab / Cyber Analysts

17. **Correlating Seized Device Extractions (UFED / Cellebrite / Oxygen):**

- When extracting WhatsApp chats, SMS, and call history from a seized phone, how do you correlate that data with telecom provider CDRs?
- How do you handle hidden/deleted contacts, encrypted vault apps, or dual-space profiles on seized phones?

18. **OSINT & Social Media Intelligence (SOCMINT):**

- How do you monitor and extract intelligence from public/private Telegram channels, Instagram handles, and dark web forums used by gang members?
- How do you connect an anonymous handle (e.g., @don_007 on Telegram) to a real-world suspect node?

Section 6: Legal Admissibility & Bharatiya Sakshya Adhiniyam (BSA 2023) For Prosecution Liaison & Senior IOs

19. BSA 2023 / Section 65B Electronic Evidence Compliance:

- What are the mandatory legal requirements for digital network graphs and AI reports to be accepted without dispute in Indian courts under the new Bharatiya Sakshya Adhiniyam (BSA 2023)?
- How do you currently prove the integrity and non-tampering of CDR files, chat logs, and digital evidence from seizure to trial?

20. Automated Hash Auditing & Court Dossier Generation:

- Do you currently maintain a digital hash log (SHA-256 / MD5) for every file ingested during an investigation?
- When preparing a final charge-sheet or remand application, what specific charts, tables, and summaries must be included to explain complex criminal networks to the Judge?

Section 7: Tooling Flaws & The "Dream Feature" For Technical Cell Operators

21. Current Tools & Frustrations:

- What commercial tools are currently used (IBM i2, indigenous CDR tools, Excel)?
- What are the top 3 biggest flaws (crashes on 500k+ rows, expensive licenses, separate tools for CDR and Bank records, no AI explanation)?

22. The Dream Feature:

- "If you could press a single button during a live kidnapping, cyber extortion, or gang investigation, what exact insight or visualization would you want on your screen within 10 seconds?"

Section 8: Field Data Collection Checklist (Sanitized Templates)

Collect?	Template / Schema Needed	Purpose in Project SHADOW-INTEL
[ ]	Blank Telecom CDR CSV / Excel Header (Airtel, Jio, Vi, BSNL)	Build accurate telecom ingestion parsers and timezone normalizers.
[ ]	Blank Tower Dump Excel Header	Configure geofencing and cell-site filtering algorithms.
[ ]	Blank 1930 / I4C Cyber Crime Complaint CSV format	Design automated financial fraud ingestion pipelines.
[ ]	Blank Bank Statement transaction export schema	Build multi-hop UPI and mule account graph algorithms.
[ ]	Sample FIR structural schema (CCTNS fields)	Train Indic NLP / NER entity extraction models.
[ ]	Sample Section 65B / BSA 2023 Evidence Certificate	Generate automated court-ready electronic evidence dossiers.